Appendix A
Differential equations

As will be evident from the models discussed in Chapter 2, many systems have input–output relationships which have to be described by differential equations. A *differential equation* is an equation involving derivatives of a function, i.e. terms such as dy/dt and d^2y/dt^2. Thus:

$$\tau\frac{dy}{dt}+y=0$$

is a differential equation. The term *ordinary differential equation* is used when there are only derivatives of one variable, e.g. we only have terms such as dy/dt and d^2y/dt^2 and not additionally dx/dt or dx^2/dt^2. The *order* of a differential equation is equal to the order of the highest derivative that appears in the equation. For example,

$$\tau\frac{dy}{dt}+y=0$$

and

$$\tau\frac{dy}{dt}+y=kx$$

are first-order ordinary differential equations since the highest derivative is dy/dt and there are derivatives of only one variable. The first of the above two equations is said to be *homogeneous* since it only contains terms involving y. Such an equation is given by a system which has no forcing input; one with a forcing input gives a non-homogeneous equation. For example, an electrical circuit containing just a charged capacitor in series with a resistor will give a homogeneous differential equation describing how the potential difference across the capacitor changes with time as the charge leaks off the capacitor; there is no external source of voltage. However, if we have a voltage source which is switched into the circuit with the capacitor and resistor then the voltage source gives a forcing input and a non-homogeneous equation results.

The equations:

$$m\frac{d^2y}{dt^2}+c\frac{dy}{dt}+ky=0$$

and

$$m\frac{d^2y}{dt^2} + c\frac{dy}{dt} + ky = F$$

are examples of a second-order differential equation since the highest derivative is d^2y/dt^2. The first of the two is homogeneous since there is no forcing input and the second is non-homogeneous with a forcing input F.

Solving a first-order differential equation

With a first-order differential equation, if the variables are separable, i.e. it is of the form:

$$\frac{dy}{dt} = f(t)$$

then we can solve such equations by integrating both sides of the equation with respect to x:

$$\int \frac{dy}{dt}\, dt = \int f(t)\, dt$$

This is equivalent to separating the variables and writing:

$$\int dt = \int f(t)\, dt$$

Consider the response of a first-order system to a step input, e.g. the response of a thermometer when inserted suddenly into a hot liquid. This sudden change is an example of a step input. In Chapter 2 the differential equation for such a change was determined as:

$$RC\frac{dT}{dt} + T = T_L$$

where T is the temperature indicated by the thermometer, T_L the temperature of the hot liquid, R the thermal resistance and C the thermal capacitance. We can solve such an equation by the technique of 'separation of variables'. Separating the variables gives:

$$\frac{1}{T_L - T}\, dT = \frac{1}{RC}\, dt$$

Integrating then gives:

$$-\ln (T_L - T) = (1/RC)t + A$$

where A is a constant. This equation can be written as:

$$T_L - T = e^A\, e^{-t/\tau} = B\, e^{-t/\tau}$$

with B being a constant and $\tau = RC$. τ is termed the *time constant* and can be considered to be the time that makes the exponential term e^{-1}. If we consider the thermometer to have been inserted into the hot liquid at

time $t = 0$ and to have been indicating the temperature T_0 at that time, then, since $e^0 = 1$, we have $B = T_L - T$. Hence the equation can be written as:

$$T = (T_0 - T_L)\, e^{-t/\tau} + T_L$$

The exponential term will die away as t increases and so gives the transient part of the response. T_L is the steady-state value that will be attained eventually.

Complementary function and particular integral

Suppose we have the first-order differential equation $dy/dt + y = 0$. Such an equation is homogeneous and, if we apply the technique of separation of the variables, has the solution $y = C\, e^{-t}$. Now suppose we have the non-homogeneous equation $dy/dt + y = 2$. If we now use the separation of variable technique we obtain the solution $y = C\, e^{-t} + 2$. Thus, its solution is the sum of the solution for the homogeneous equation plus another term. The solution of the homogeneous differential equation is called the *complementary function* and, when there is a forcing input with that differential equation, the term added to it for the non-homogeneous solution is called the *particular integral*. We can, easily, obtain a particular integral by assuming it will be of the same form as the forcing input. Thus if this is a constant then we try $y = A$, if of the form $a + bx + cx^2 + \ldots$ then $y = A + Bx + Cx^2 + \ldots$ is tried, if an exponential then $y = A\, e^{kx}$ is tried, if a sine or cosine then $y = A \sin \omega x + B \cos \omega x$ is tried.

As an illustration, consider the system involving solved in the previous section of a thermometer being inserted into a hot liquid and the relationship being given by:

$$RC\frac{dT}{dt} + T = T_L$$

The homogeneous form of this equation is:

$$RC\frac{dT}{dt} + T = 0$$

We can solve this equation by the technique of the separation of variables to give the complementary function:

$$\frac{1}{T}\, dT = -\frac{1}{RC}\, dt$$

Integrating then gives:

$$-\ln T = -(1/RC)t + A$$

where A is a constant. This equation can be written as:

$$-T = e^A\, e^{-t/\tau} = B\, e^{-t/\tau}$$

$$T = -B\,e^{-t/\tau}$$

Now consider the particular integral for the non-homogeneous equation and, because the forcing input is a constant, we try $T = C$. Since $dT/dt = 0$ then the substituting values in the differential equation gives:

$$0 + C = T_L$$

Thus the particular solution is $T = T_L$. Hence the full solution is the sum of the complementary solution and the particular integral and so:

$$T = -B\,e^{-t/\tau} + T_L$$

If we consider the thermometer to have been inserted into the hot liquid at time $t = 0$ and to have been indicating the temperature T_0 at that time, then, since $e^0 = 1$, we have $B = T_L - T$. Hence, as before, the equation can be written as:

$$T = (T_0 - T_L)\,e^{-t/\tau} + T_L$$

As a further illustration, consider the thermometer, in equilibrium in a liquid at temperature T_0, when the temperature of the liquid is increased at a constant rate a, i.e. a so-called ramp input. The temperature will vary with time so that after a time t it is $at + T_0$. This is then the forcing input to the system and so we have the differential equation:

$$RC\frac{dT}{dt} + T = at + T_L$$

The homogeneous form of this equation is:

$$RC\frac{dT}{dt} + T = 0$$

and, as before, the complementary solution is:

$$T = -B\,e^{-t/\tau}$$

For the particular integral we try a solution of the form $T = D + Et$. Since $dT/dt = E$, substituting in the non-homogeneous differential equation gives:

$$RCE + D + Et = aT + T_0$$

Equating all the coefficients of the t terms gives $E = a$ and equating all the non-t terms gives $RCE + D = T_0$ and so $D = T_0 - RCa$. Thus the particular integral is:

$$T = T_0 - RCa + at$$

and so the full solution, with $\tau = RC$, is:

$$T = -B\,e^{-t/\tau} + T_0 - \tau a + at$$

When $t = 0$ we have $T = T_0$ and so $B = -\tau a$. Hence we have:

$$T = \tau a\,e^{-t/\tau} + T_0 - \tau a + at$$

Solving a second-order differential equation

We can use the technique of finding the complementary function and the particular integral to obtain the solution of a second-order differential equation. Consider a spring–damper–mass system of the form shown in Figure 2.8. The differential equation for the displacement y of the mass when subject to step input at time $t = 0$ of a force F is:

$$m\frac{d^2y}{dt^2} + c\frac{dy}{dt} + ky = F$$

In the absence of damping and the force F we have the homogeneous differential equation :

$$m\frac{d^2y}{dt^2} + ky = 0$$

This describes an oscillation which has an acceleration d^2y/dt^2 which is proportional to $-y$ and is a description of simple harmonic motion; we have a mass on a spring allowed to freely oscillate without any damping. Simple harmonic motion has a displacement $y = A \sin \omega t$. If we substitute this into the differential equation we obtain:

$$-mA\omega^2 \sin \omega t + kA\omega \sin \omega t = 0$$

and so $\omega = \sqrt{(k/m)}$. This is termed the *natural angular frequency* ω_n. If we define a constant called the *damping ratio* of:

$$\zeta = \frac{c}{2\sqrt{mk}}$$

then we can write the differential equation as:

$$\frac{1}{\omega_n^2}\frac{d^2y}{dt^2} + \frac{2\zeta}{\omega_n}\frac{dy}{dt} + y = \frac{F}{k}$$

This differential equation can be solved by the method of determining the complementary function and the particular integral. For the homogeneous form of the differential equation, i.e. the equation with zero input, we have:

$$\frac{1}{\omega_n^2}\frac{d^2y}{dt^2} + \frac{2\zeta}{\omega_n}\frac{dy}{dt} + y = 0$$

We can try a solution of the form $y = A\,e^{st}$. This, when substituted, gives:

$$\frac{1}{\omega_n^2}s^2 + \frac{2\zeta}{\omega_n}s + 1 = 0$$

$$s^2 + 2\omega_n\zeta s + \omega_n^2 = 0$$

This equation has the roots:

$$s = \frac{-2\omega_n\zeta \pm \sqrt{4\omega_n^2\zeta^2 - 4\omega_n^2}}{2} = -\omega_n\zeta \pm \omega_n\sqrt{\zeta^2 - 1}$$

When we have:

1 *Damping ratio between 0 and 1*
There are two complex roots:

$$s = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$$

If we let:

$$\omega = \omega_n\sqrt{1-\zeta^2}$$

we can write:

$$s = -\zeta\omega_n \pm j\omega$$

Thus:

$$y = A\,e^{(-\zeta\omega_n + j\omega)t} + B\,e^{-\zeta(\omega_n - j\omega)t}$$

$$= e^{-\zeta\omega_n t}(A\,e^{j\psi t} + B\,e^{-j\omega t})$$

Using Euler's equation, we can write this as:

$$y = e^{-\zeta\omega_n t}(P\cos\omega t + Q\sin\omega t)$$

This can be written in an alternative form. If we consider a right-angled triangle with angle ϕ with P and Q being opposite sides of the triangle (Figure ApA.1) then $\sin\phi = P/\sqrt{(P^2 + Q^2)}$ and $\cos\phi = P/\sqrt{(P^2 + Q^2)}$. Hence, using the relationship $\sin(\omega t + \phi) = \sin\omega t\cos\phi + \cos\omega t\sin\phi$, we can write:

$$y = C\,e^{-\zeta\omega_n t}\sin(\omega t + \phi)$$

where C is a constant and ϕ a phase difference. This describes a damped sinusoidal oscillation. Such a motion is said to be *under damped.*

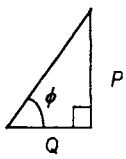

Figure ApA.1 *Angle* ϕ

2 *Damping ratio equal to 1*
This gives two equal roots $s_1 = s_2 = -\omega_n$. and the solution:

$$y = (At + B)\,e^{-\omega_n t}$$

where A and B are constants. This describes an exponential decay with no oscillations. Such a motion is said to be *critically damped.*

3 *Damping ratio greater than 1*
This gives two real roots:

$$s_1 = -\omega_n\zeta + \omega_n\sqrt{\zeta^2 - 1}$$

$$s_2 = -\omega_n\zeta - \omega_n\sqrt{\zeta^2 - 1}$$

and hence:

$$y = A\,e^{s_1 t} + B\,e^{s_2 t}$$

where A and B are constants. This describes an exponential decay taking longer to reach the steady-state value than the critically damped case. It is said to be *over damped.*

The above analysis has given the complementary functions for the second-order differential equation. For the particular integral, in this case where we have a step input of size F, we can try the particular integral $x = A$. Substituting this in the differential equation gives $A = F/k$ and thus the particular integral is $y = F/k$. Thus the solutions to the differential equation are:

1 *Damping ratio between 0 and 1, i.e. under damped*

$$y = C\,e^{-\zeta\omega_n t}\sin(\omega t + \phi) + F/k$$

2 *Damping ratio equal to 1, i.e. critically damped*

$$y = (At + B)\,e^{-\omega_n t} + F/k$$

3 *Damping ratio greater than 1, i.e. over damped*

$$y = A\,e^{s_1 t} + B\,e^{s_2 t} + F/k$$

In all cases, as t tends to infinite then y tends to the value F/k. Thus the steady-state value is F/k.